
BEVY_ZEROVERSE: AUDITABLE PROCEDURAL INDOOR DATA FOR MULTI-VIEW RECONSTRUCTION

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ABSTRACT

We describe an open-source synthetic data engine combining primitive-object scenes with furnished procedural interiors, calibrated multi-view cameras, optional human motion, and aligned geometric annotations. Indoor generation samples continuous architectural, furnishing, appearance, lighting and camera parameters under geometric constraints. A generator-20 camera evaluation covers 2,048 consecutive layouts and 512 rendered rooms, including all-camera co-visibility, spatial spread and trajectory diversity. We retain the separate generator-18 evaluation of 1024 layouts and 64 rendered rooms for appearance distributions, image-embedding variance and completed-capture throughput. Its measured CPU optimization reduces matched room-capture time by 33.6%, without reducing texture resolution, mesh detail or lighting samples. Published registry dependencies are measured separately from local graphics-library patches. These are generator and dataset diagnostics: photographic realism, unlimited-process stability and downstream pretraining gains are not established by this study.

1 Purpose and scope

Sparse-view reconstruction benefits from observations of shared surfaces: cameras that see unrelated objects offer little direct correspondence supervision. LRM-Zero demonstrates the potential of procedurally synthesized object data [1]; MVSplat uses cross-view matching to estimate geometry [2]. Our contribution is an inspectable data-generation system and its bounded evaluation, rather than a trained reconstruction model or a reconstruction accuracy result.

Version 0.24 retains the existing Object, CornellCube, Room, SemanticRoom and Human scene modes, and exposes furnished interiors separately as `ProceduralIndoor`. The runtime uses Bevy 0.19.1, optional Burn 0.21 inference, and published AnnyBody/ARDY integrations. Office, conference, lounge, classroom, library and related activity priors influence a compositional program; they do not select complete meshes or a finite collection of room templates. Most architecture, furniture, plants and clutter require neither mesh repositories nor photographic texture collections. AnnyBody human meshes and optional motion/embedding models are explicit pretrained dependencies.

2 Procedural scene program

A scene is generated as $S = G(s, \theta)$, where s seeds independent random streams and θ contains user-controlled densities, activity, quality, camera and motion policies. The serialized manifest records the sampled program, object transforms and seeds, people, photometry and camera paths. A separate generator version and capture-engine identity invalidate incompatible resummptions. Determinism means reproducing the same program and settings with compatible assets and engine versions; bit-identical rasterization across GPU vendors is not promised.

2.1 Architecture, objects and appearance

Room area and aspect are sampled logarithmically; width and depth are clamped to 5.6–21 m and height spans 2.65–4.8 m. Furnishing zones have continuous bounds, orientation, aisle width, occupancy and mixed activity weights. Partitions

sample portal width, thickness, glazing fraction, mullion spacing and optional transoms. Pillars, ceiling relief, paneling, reveals, trim and shelves add geometric structure. One to three exterior faces independently receive glazing; the fourth face retains the neighboring-room connection. Window opening dimensions, sill/head heights, bay widths, frames, inset depth and shades vary continuously. Rough openings are subtracted from the solid wall geometry; decorative surfaces respect these openings. The exterior envelope remains rectangular: these are varied rectilinear interiors, not a general architectural CAD system.

Furniture is assembled from material-specific parts. Tables vary outline, taper, thickness and leg/pedestal construction; chairs vary seat shell, curvature, back proportions, recline, arms, base and headrest. Sectional seating, shelving, potted plants, outlets, switches, ceiling lights, whiteboards, displays and tabletop objects extend the semantic inventory. Placement uses support relationships, clearance checks, zone-dependent furnishing fields and bounded rejection. Object yaw and arrangement are sampled rather than restricted to orthogonal grids. Collision proxies approximate detailed meshes; geometric validity is checked separately from physical plausibility.

Surfaces use 35 role-specific base material slots with procedural albedo, normal and roughness maps, metric UV scales and mip chains. Recipes vary spatial frequency, directional grain, textile structure, mineral veining, relief, wear, color and roughness. Object finish variants preserve their underlying maps while varying appearance; screens, boards and printed pages have independently seeded content. Only referenced finish variants are built during runtime preparation. Glass recipes vary roughness, refractive index, tint, attenuation and thickness. Direct lights vary placement, intensity, color temperature, circuit activity and direction; exposure, daylight and sky radiance are independently recorded. Low-light samples are retained in evaluation.

People retain AnnyBody geometry with continuous phenotype/pose parameters, fitted garments, surface hair and facial details. Static people remain the default. Opt-in ARDY motion combines seeded procedural text sequences with object-aware waypoint planning and scene-time playback. Requesting static scenes does not initialize motion inference. Motion models use the upstream loaders and caches; cached prompt text does not replace the motion seed. Clothing, hair, joint limits and collision handling are approximations, particularly during complex motion.

2.2 Multi-view sampling and training defaults

Camera centers and complete trajectories are constrained to the primary furnished room by default. The sampler varies focal length, position, target, roll and path shape, checks obstacle clearance and prefers visible content. Static capture uses one timestep; optional trajectories support repeated calibrated observations. A reference camera anchors a connected set of views. For each other camera and five trajectory times, we require

$$\min\{o_{0 \rightarrow i}(t), o_{i \rightarrow 0}(t)\} \geq \tau, \quad b_{\min} \leq \|c_i(t) - c_0(t)\| \leq b_{\max},$$

where o estimates first-surface overlap using coarse visibility rays and object/architecture proxies. Overlap constrains reference edges, not every camera pair. Generator 20 additionally enforces $b_{\min}$ between every camera pair at 33 synchronized path samples. For three or more views, the horizontal camera-center covariance must have minor/major standard-deviation ratio at least 0.25 at those times. Paths independently vary heading, distance and curvature around the reference route; moving pairs require normalized RMS displacement difference of at least 0.15 after removing their starting offsets. Controls permit explicit rigid or collinear rigs. The finite proposal budget fails explicitly rather than silently weakening thresholds. Full-path swept collision checks remain; sampled spread and overlap are not continuous-time or pixel-level guarantees.

3 Capture and annotation contract

Capture begins only after scene preparation, assets, shader pipelines, lighting and requested human trajectories are ready. A bounded asynchronous readback protocol associates outputs with the requested scene, camera, time and mode; it does not infer readiness from an arbitrary number of frames. Geometric channels share an RGBA32Float ground-truth raster attachment. RGB uses Bevy’s PBR pipeline, shadow maps, tone mapping and optional indirect-lighting approximation. Float32 storage does not imply that all RGB or lighting computations have float32 intermediate precision: HDR color and irradiance volumes include float16 stages.

Headless generation supports chunked safetensors or per-sample files, compression, resumable identity checks, metrics and bounded worker replacement. RGB and numeric channels have separate encoding contracts. Native generation performs numeric temporal export and O-Voxel export. The WebGPU viewer supports indoor construction, inspection and optional motion inference, using the same manifest and geometry modules; Portable quality reduces unsupported or costly rendering features. Browser velocity visualization is not a qualification of native dataset flow export. The current Wasm path omits native baked diffuse GI and uses its platform lighting fallback.

Indoor training configuration	Default and rationale
Capture cameras / timesteps	Four / one; synchronized views of static geometry
Shared-view policy	$\tau = 0.35$, all-pair separation ≥ 0.25 m, reference baseline ≤ 3 m; explicit <code>multiview:null</code> enables independent views
Group diversity	Horizontal spread ratio ≥ 0.25 for 3+ views; moving-pair normalized RMS difference ≥ 0.15
Room/activity/density	Primary room, Mixed activity, furnishing density 0.65, human density 0.25
Camera path policy	0.03–8 m allowed path length, 0.45 long-path proposal fraction; collision rejection still applies
Human motion	Disabled; static people remain, no ARDY initialization
Native lighting	Auto quality, shadows and approximate baked diffuse GI, 256 rays/probe
Dataset CLI workers / chunks	One process, 16 rooms/chunk; explicit overrides retained
Finite worker lifetime	256 scenes/child for finite indoor jobs with process workers
Default export	RGB and calibration; geometric channels are opt-in

Table 1: Defaults conditional on the procedural-indoor dataset CLI scene. Other scene modes retain their defaults. These are an engineering starting point, not an optimum established by downstream training. Library callers explicitly choose camera and worker counts.

Output	Meaning and qualification
Color	Display RGB; lossless float storage or explicitly chosen codecs. Separate linear-HDR export supports renderer comparisons.
Depth	Colorized, normalized or metric linear depth; axial/ray conventions; calibration records the selection.
Normal / position	View-space normals encoded as $(n + 1)/2$; world positions affine-normalized by the exported AABB, without clamping. First surface, float32 geometric path.
Semantic	Shared class vocabulary, aligned with depth/normal/position, including ceiling light fixtures.
Optical flow	Forward displacement $(\Delta u, \Delta v)$ in pixels, with correspondence-valid and target-visible masks. Terminal time has no forward correspondence.
Motion vectors	The same exported displacement divided by image width/height. Viewer velocity colors are previews, not these numeric annotations.
Co-visibility	Optional same-time, per-pixel membership in other capture cameras. A uint16 mask assigns one bit per camera (up to 16), with a separate source-valid mask and a consistent additive RGB legend; the source camera and editor are excluded.
Calibration / instances	Camera intrinsics, poses and normalized trajectory time, scene manifest, semantic instance oriented boxes, person joints, and primary-room annotation AABB.
O-Voxel	Primary-room architecture/objects only; excludes neighboring/outside geometry. Requires one timestep and disabled human motion; incompatible requests are rejected.

Table 2: Eight currently supported render modes plus structured annotations. The population experiment below uses five modes; co-visibility was added subsequently. Glass is a first geometric surface: annotations do not follow refracted or reflected light paths. Flow transports surface correspondence, including supported human deformation, rather than estimating photometric image motion.

4 Evaluation protocol and results

The evaluation artifacts and commands are in `docs/generation_v18.md`; machine-readable results generate this paper’s tables. We audit 1024 consecutive seeds starting at 24000 at density 0.65, human density 0.25 and four cameras, with no failed-seed replacement. The first 64 seeds are rendered at 384×288 with RGB, depth, normal, position and semantic outputs. Every scene contributes to the reports, including dark rooms and weak views. Geometric audits and rendered-camera checks answer different questions and are reported separately.

4.1 Diversity and effective variance

The full export includes 503 numeric distributions, 401 of which have nonzero observed variance, 33 categorical distributions, zero-preserving per-class object counts, placement heatmaps, camera intrinsic/extrinsic distributions and joint histograms. These series are correlated projections of the program, not independent degrees of freedom.

Audited rooms / cameras	1024 / 4096
Room area (m ²)	38.01–267.28 (mean 117.84)
Main-interior chairs per scene (zeros included)	1.00–26.00 (mean 9.09)
People per scene	0.00–9.00 (mean 2.32)
Exterior window walls / near full-height rooms	1–3 / 337
Solar illuminance (lux)	0.10–99862.07 (mean 20026.91)
Vertical field of view (degrees)	28.07–106.97 (mean 65.04)
Observed / entropy-effective activity categories	10 / 9.96
Quantized occupancy-signature estimate	1018.8

Table 3: Consecutive layout population. Occupancy uses HyperLogLog with 4096 registers (approximately 1.63% relative standard error), 12×12 class occupancy and 10 cm partition quantization; random seeds and appearance are excluded.

Histograms include their denominators and units; per-object style frequencies must not be interpreted as per-room frequencies.

For a categorical distribution p , we report the entropy-effective count $\exp(-\sum_j p_j \log p_j)$. For centered feature eigenvalues λ_j , we use $q_j = \lambda_j / \sum_k \lambda_k$ and entropy-effective rank $r_{\text{eff}} = \exp(-\sum_j q_j \log q_j)$. Constant features have rank zero. We additionally measure normalized SigLIP2 base image embeddings [3], with frozen model/shard identity and declared resize/normalization. Nearest-neighbor distances exclude all views and times from the same scene. Scene-level variance uses one normalized mean embedding per room, so four cameras do not count as four independent scenes.

Rooms / images / embedding dimensions	64 / 256 / 768
Cross-room nearest cosine distance: min / median / max	0.0394 / 0.0628 / 0.1832
View fraction with cross-room nearest distance below 0.02	0.00%
Scene-centroid entropy-effective rank	32.29
Dimensions explaining 90% of centroid variance	34

Table 4: Frozen SigLIP2 base-patch16-224 audit. Direct bilinear RGB resize, mean/std 0.5, L2-normalized embeddings. Thresholds are exploratory; no real-data baseline or training-utility claim.

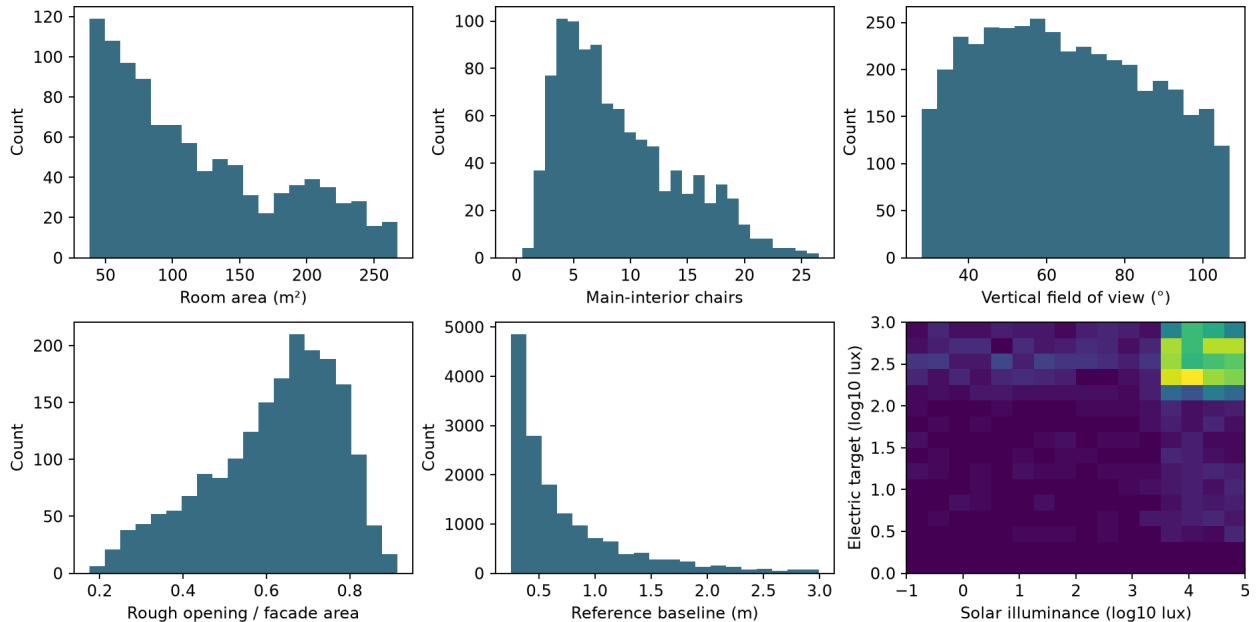


Figure 1: Observed variation in consecutive generated rooms. Shape, furnishing, illumination and camera metrics complement image-embedding diagnostics; none alone measures photographic realism.

A seed’s bit width does not establish dataset diversity. Continuous parameter ranges have no meaningful finite cardinality before quantization; topology, placement constraints, correlations and rendering further collapse distinguishable outcomes. The occupancy-signature estimate in Table 3 ignores seeds and textures and uses a declared spatial quantization. It is a bounded geometric redundancy diagnostic, not proof of ten million useful examples. Embedding rank and nearest distances also depend on encoder, population size, viewpoint policy and illumination. The study does not identify a universal acceptable distance or train a downstream model.

4.2 Multi-view and geometric correctness

Semantic classes/view: minimum / mean (at least 32 pixels)	7 / 11.38
Occupied primary rooms with no person label	0 / 45
Occupied non-neighbor rooms with no person label	14 / 60
Rendered reference edges / rooms	192 / 64
Worst-pair overlap per room: min / mean	0.344 / 0.474
Rendered pairs below the 0.35 proxy requirement	3
Maximum per-view depth/position p99 error (m)	2.86e-06
Maximum per-view reprojection p99 error (pixels)	0.00262

Table 5: Actual geometry audit at the captured timestep. Visibility requires depth agreement within 0.01 m plus 0.002 times depth; nearest-pixel raster boundaries and thin geometry differ from proxy rays.

Rendered overlap is computed by depth unprojection, world-to-camera reprojection, frustum tests and a declared depth-consistency tolerance. We retain failures below the requested proxy threshold. Indoor annotation checks also compare depth with world-position reconstruction, reprojection and finite unit normals. The population audit found 0 rejected/invalid seeds; this finite result is not an exhaustive correctness proof. Exact configuration, capture engine, time and camera identity accompany all checks.



Figure 2: Reference-camera RGB for the first eight consecutive rendered seeds, 24000–24007, without aesthetic filtering. The retained cohort includes dark scenes, approximate glass and visibly procedural human/material appearance.

4.3 Generation performance

Timing uses completed samples, including asynchronous construction, lighting, rendering and readback. It excludes image/dataset encoding, validation and report writing unless explicitly labeled end-to-end. Each main benchmark runs 40 consecutive rooms, discards eight warmup rooms and captures four views per room at 320×240 , one timestep, with static humans at density 0.25. Five-channel runs request RGB/depth/position/normal/semantic. Hardware is an NVIDIA RTX PRO 6000 Blackwell workstation GPU, Vulkan driver 610.43.02 on Linux. The benchmark records adapter, CPU, compiler, process memory, GPU timesteps and matching-PID NVML telemetry. Other desktop GPU activity is recorded, not attributed to the generator.

baseline	2.84 views/s; 1.348/2.364 s
shared geometry	2.68 views/s; 1.382/2.540 s
optimized	3.31 views/s; 1.073/2.078 s
parallel	4.28 views/s; 0.761/1.775 s
registry grouped	3.58 views/s; 0.892/2.642 s
registry rgb	4.32 views/s; 0.773/1.776 s

Table 6: Completed views per second; room wall-time median/p95. Baseline/shared geometry/optimized/parallel are local-patch runs with identical independent-camera settings. Registry rows use the normalized published dependency graph and the new grouped-camera default. See the evidence for modes, startup, memory and stage timings.

Profiling separates layout, camera planning, materials, geometry, tangent conversion and asset insertion. The old “preparation” field measured only the regeneration request, concealing the asynchronous work. The revised schema records both the request and actual CPU stages. Geometry is constructed once for transport and rendering; unused material variants are skipped; independent MikkTSpace mesh conversions use a bounded four-thread native pool. The development profile now optimizes the procedural crate at level 3 as well as dependencies. Wasm retains cooperative serial conversion. No mesh-detail, texture-resolution, GI-ray or view-count reduction is used in the comparison. The intermediate shared-geometry-only trial did not improve throughput and is retained in the evidence.

The before/after comparison changes both source preparation and the root development optimization level (1 to 3); it is not a source-only ablation or a release-LTO benchmark. Registry measurements compile the normalized crate archive without checkout-only wgpu patches. They should be used for published-package expectations. GPU timestamps are asynchronous diagnostic samples, and NVML active-process samples are not continuous occupancy measurements. Short runs cannot prove unlimited memory stability; finite child lifetimes remain part of the large-dataset operating policy. Throughput excludes neural motion generation and depends strongly on scene size, view count, resolution, modes and hardware.

5 Expanded camera and co-visibility evaluation

We compare generators 19 and 20 on 2,048 consecutive layout seeds (24000–26047), with four cameras, furnishing density 0.65 and static human density 0.25. Object, people and architecture CSV hashes agree exactly. All new layouts pass geometric, swept-path and sampled group checks. The new export contains 507 numeric series, 405 with observed nonzero variance; these remain correlated diagnostics, not independent degrees of freedom. The source reports and commands are retained in docs/camera_evaluation_v20.md.

Camera-group statistic	Generator 19	Generator 20
Near-collinear starting groups (spread < 0.25)	626	0
Median starting spread	0.351	0.474
Median minimum pair separation (m)	0.306	0.431
Median minimum relative motion	approximately zero	0.341

Table 7: Matched layout comparison. Legacy CSVs expose endpoint checks; new path metrics use 33 synchronized samples. Generator 19 translated the same reference path for all views.

The rendered cohort contains 512 consecutive rooms (24000–24511), four 320×240 cameras and times 0, 0.5, 1: 6,144 views in 4 completed process chunks. Native lights and shadows remain enabled; baked GI is disabled for this geometric audit. No dark, weak or low-overlap view is discarded. Shared-depth reference overlap over 4,608 pair/time observations averages 53.4%, with minimum 20.5%. 245 observations miss the 35% proxy target; 0 rooms contain a reference pair below 10%.

We additionally evaluate all 18,432 directed camera-pair/time observations. For each valid source pixel we form a membership mask of other cameras passing rendered-depth reprojection, excluding the source bit. Across 471,859,197 valid source-pixel observations, the fractions seen by zero, one, two and three other cameras are 15.7%, 20.7%, 25.6% and 38.0%, respectively. Observations repeat surfaces across views and times; these are not counts of unique 3D points. 0 of 1,536 synchronized sets have a disconnected graph when edges require 10% bidirectional overlap.

This independent diagnostic uses nearest-pixel depth agreement within $0.01\text{ m} + 0.002z$. It is deliberately separate from the production GPU co-visibility annotation’s symmetric tangent-plane test described below; its masks must not

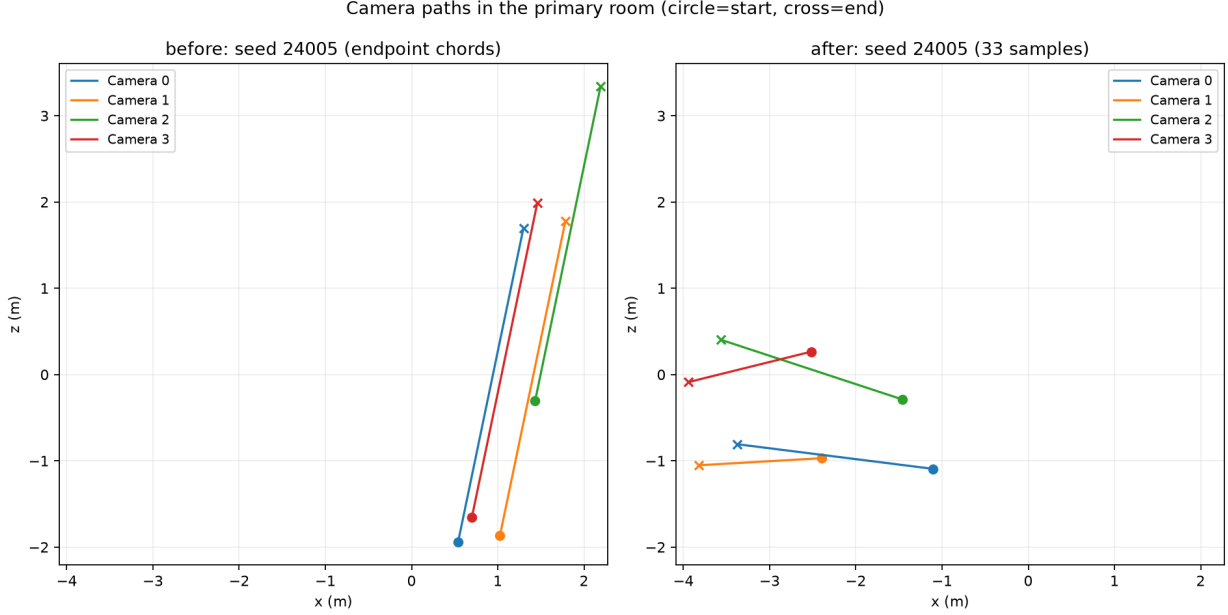


Figure 3: Seed 24005 before and after the group/path diversity constraints. Starts are circles, endpoints crosses.

be interpreted as identical annotation values. Both treat glass as an opaque geometric surface. Gallery mask exports retain and validate the production annotation contract.

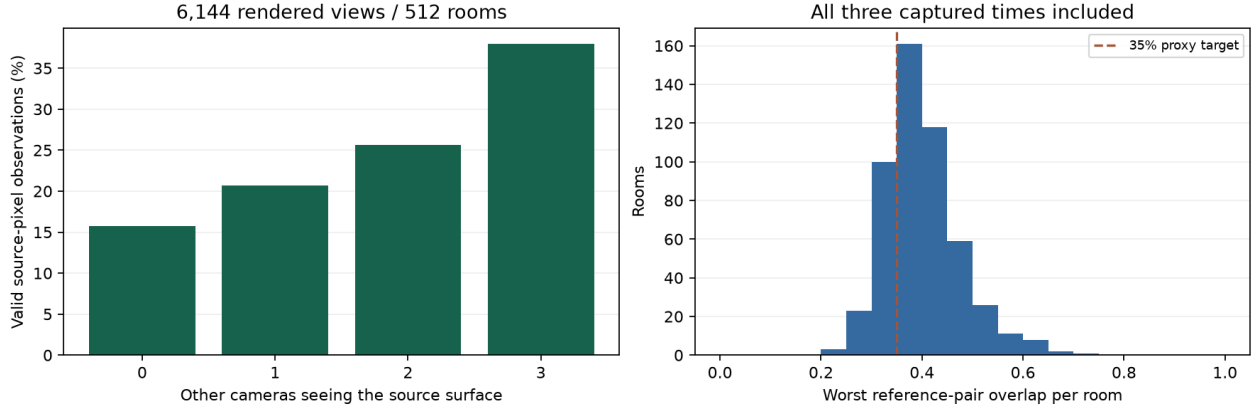


Figure 4: Expanded co-visibility evaluation: source-pixel membership cardinality and per-room worst reference overlap. All captured times and failures are retained.

The same captures expose quality tails: 0 rooms contain a view with at most two visible semantic classes; 0 of 357 rooms containing primary-room people have no person pixels across the captured views. These diagnostics neither establish photographic realism nor repeat the historical embedding and throughput experiments. The largest depth/position p99 disagreement is 0.000004 m. Four bounded processes do not establish unlimited-process memory stability.

6 Subsequent gallery and viewer update

The accompanying project page presents selected generator-20 renders separately from the population evaluations. Two example rooms (seeds 24005 and 24000) were captured at 768×480 , with four synchronized cameras, three normalized trajectory times and all eight render modes. All mode switches reuse these exact scene/camera/time identities. The native CLI used Auto quality, static human density 0.25 and 1024 indirect-lighting rays per probe. Depth/position reconstruction and camera reprojection checks validate alignment before generating website previews.

Display mappings for depth, normals, positions and temporal vectors are declared alongside the visualizations; they do not replace the lossless numeric exports.

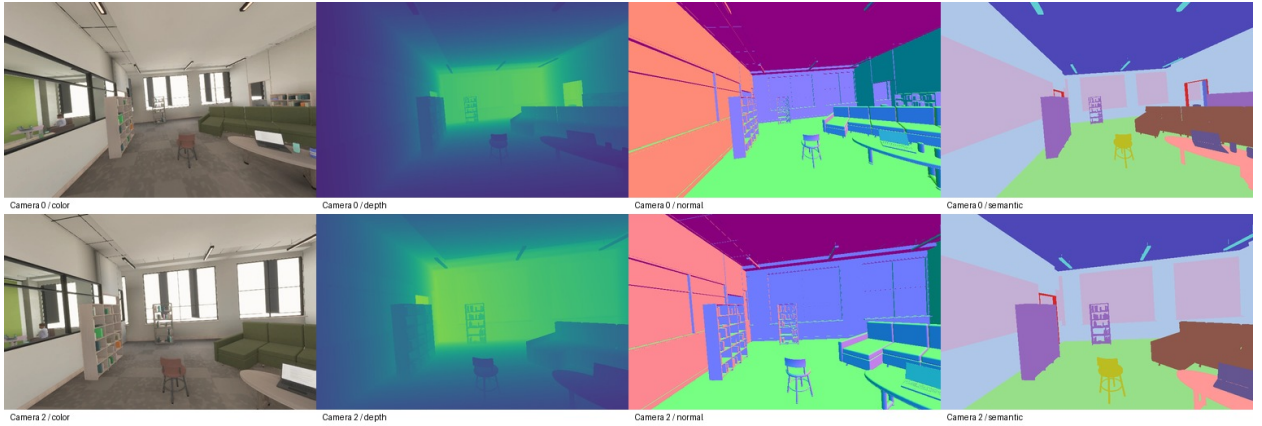


Figure 5: Selected generator-20 room, seed 24005: RGB, metric-depth visualization, view-space normals and semantic labels for cameras 0 and 2 at the same time. Camera selection and pixel-aligned annotation comparisons are interactive on the project page. These illustrations are separate from the consecutive population evaluation.

Co-visibility shares the float32 surface raster used by the other geometric annotations. A GPU pass projects each source surface into the other cameras and checks frustum bounds and symmetric local tangent-plane agreement. The tolerance is $\max(1 \text{ mm}, 10^{-4}d)$ at target depth d . The exact membership mask is separate from its additive color preview and source-valid mask; a valid but unshared surface differs from background even though both have a zero membership mask. This measures raster surface visibility, with the same annotation-opaque glass policy, rather than reflected or refracted RGB content.

Generator 19 also replaces the squared clothing chest offset with smooth body-derived cross-sections, localized folds and fitted garment details. Eight matched adult wardrobes were visually reviewed before and after the change. Backless stool variants were present in 89 of 128 audited rooms (201 of 1198 seats). Viewer pose joints can show through people while environment depth occludes them; editor projections survive regeneration. ARDY controls expose diffusion steps, history length, guidance and path conditioning, and the prompt program limits optional style modifiers. Local bone translations are interpolated continuously. The project page’s separate motion example uses seed 13, two accepted trajectories and two cameras, rendered as 120 frames at the model’s 20 Hz rate. It is a bounded demonstration, not a motion-quality benchmark. Native and WebGPU validation and remaining qualifications are recorded in docs/viewer_quality_v19.md.

7 Limitations and remaining evidence

Procedural correctness, broad parameter variation and efficient capture do not establish photographic realism. Hair is a surface groom, skin lacks a full subsurface model, some furnishings remain sparse, object motion uses approximate collision proxies, and diffuse GI uses probes and simplified transport. Glass and lighting are approximations rather than full path tracing. A retained earlier matched-scene Cycles study (generator 14, four unoccupied rooms, eight views, independent raw 8192-sample references) found coarse radiance relative errors of 9.0–98.7%. That historical result is not a requalification of every generator-18 scene and explicitly includes the difficult dark example. Reference convergence and visual plausibility must be evaluated independently.

The current study is a generator audit and a small capture cohort, not a ten-million-scene run. It does not establish long-run resident-memory bounds, balanced coverage of all rare architectural combinations, collision-free arbitrary motion, suitability for every reconstruction loss, or transfer to real photographs. Shared-view defaults deliberately correlate cameras within a room; train/validation splits should therefore be made by room seed, not by image. A downstream assessment should freeze a training budget and real evaluation set, compare matched data quantities and camera policies, and measure reconstruction/calibration quality alongside redundancy. The manifests and metrics make that experiment reproducible without claiming its outcome in advance.

References

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